

SATR-SSM: SALIENCY-AWARE TOPOLOGY-ROUTED STATE SPACE MODELING FOR OBJECT DETECTION

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ABSTRACT

Visual state space models unfold two-dimensional features into one-dimensional sequences, but existing scanning schemes struggle to align state propagation with the spatial structure of the input. We propose SATR-SSM, a Saliency-Aware Topology-Routed State Space Model for object detection that comprises Multi-Scale Gaussian Attention Modulation (MGAM), Macro-Topology Routed State Propagation (MTRP), and a Macro-Topology Cross-Path Gate (MTCP). MGAM enhances salient responses across multiple scales. MTRP learns one-to-one macro-region visitation orders for different channel groups, preserves every intra-region position, and restores the spatial layout after group-wise Mamba propagation. MTCP generates a shared structural gate to calibrate the multi-path outputs. With 3.72M parameters, SATR-SSM achieves an mAP₅₀ of 0.706 and an mAP_{50:95} of 0.435 on M3FD, and an mAP₅₀ of 0.655 and an mAP_{50:95} of 0.422 on MFAD, using visible-light data. The paper and pretrained weights are available at <https://github.com/TingFeng36/SATR-SSM>.

Index Terms— visual state space model, macro-topology routing, dynamic scanning, object detection

1. INTRODUCTION

State space models enable near-linear-time sequence modeling for long-range visual dependencies. Mamba uses input-dependent selective state updates [1], and Mamba-2 unifies structured state propagation and attention through state space duality [2]. Vision Mamba, VMamba, LocalMamba, and Mamba-ND extend these models to two-dimensional representations [3, 4, 5, 6]. Unlike text sequences, a two-dimensional feature map has no unique one-dimensional traversal order. Linearization directly determines spatial adjacency along the recurrent trajectory, making sequence organization a fundamental modeling issue in visual SSMs. This sensitivity to traversal order is especially critical for object detection, where reliable localization de-

pends on preserving spatial relationships among targets, boundaries, and contextual cues across scales.

Existing methods use predefined directional scans [4], deformable or dynamic scanning [7, 8], or learnable traversal orders [9, 10]. Other visual SSMs explore sparse scanning, hybrid modeling, and state aggregation [11, 12, 13, 14, 15, 16], while MambaOut reassesses the need for Mamba/SSM mixers in vision [17]. Mamba YOLO, Mamba-ViT, and AKCMamba-YOLO demonstrate the feasibility of Mamba-based models for object detection [18, 19, 20], while Switch-UMamba and GroupMamba employ group-wise or multi-path propagation [21, 22]. Nevertheless, existing methods often tie the scale used to determine traversal orders to the scale at which state propagation is performed. Moreover, these approaches may not guarantee strict one-to-one region visitation.

We therefore aim to establish content-driven recurrent adjacency at the region level within a compact routing space while retaining all fine-grained states. SATR-SSM predicts a macro-region permutation per channel group, defining content-adaptive inter-region adjacency. After reordering, all intra-region positions form a full-resolution Mamba sequence within each group. A shared routing network with group-specific predictor heads produces one routing assignment for each group, enabling group-specific spatial organization.

Our contributions are:

- 1) We propose MTRP to reorganize recurrent adjacency through input-conditioned, group-specific one-to-one macro-region permutations while retaining all spatial positions for full-resolution state propagation.
- 2) We propose MGAM to enhance multi-scale channel and spatial saliency before macro-level routing.
- 3) We propose MTCP to extract shared pre-routing macro structure and generate channel-wise gates for calibrating restored MTRP outputs.
- 4) We construct a four-stage SATR-SSM backbone and combine it with a standard PAN-FPN neck and detection head. The backbone and complete detector contain 2.50M and 3.72M parameters, respectively. Experiments on M3FD and MFAD validate the effectiveness of the overall design.

2. METHOD

2.1. Overall Architecture

As shown in Fig. 1, the SATR-SSM backbone comprises a two-layer convolutional stem, four hierarchical SATR stages, and a Spatial

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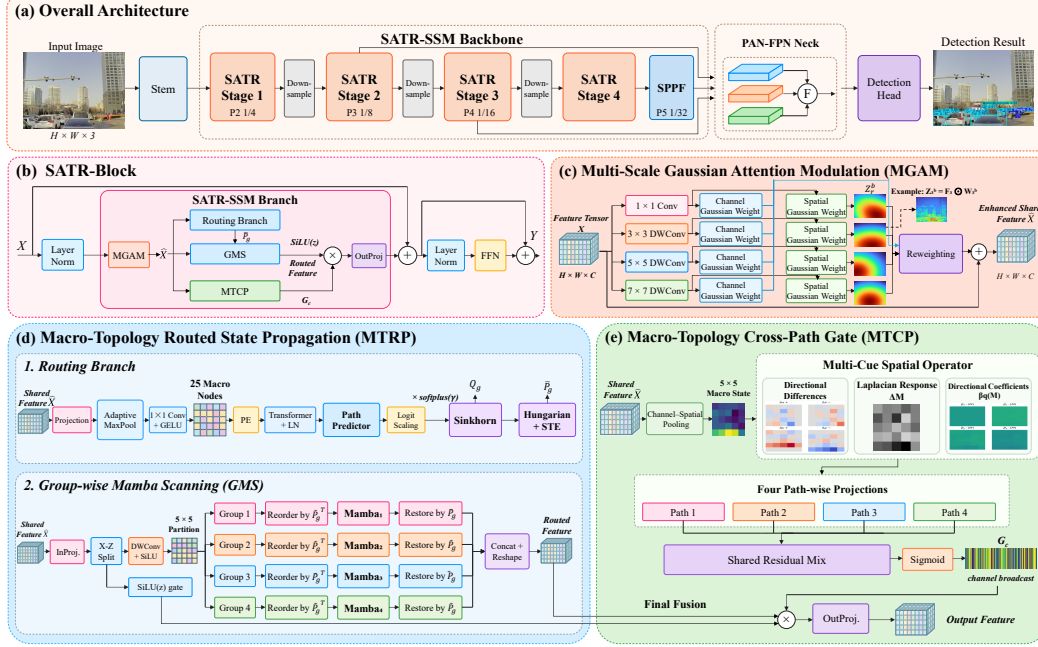


Fig. 1. Overall architecture and module flow of SATR-SSM. MTRP permutes G^2 macro regions while retaining intra-region positions; $\bar{P}_g^T$ and $\bar{P}_g$ denote reordering and restoration. Blue links from MGAM indicate channel modulation.

Pyramid Pooling–Fast (SPPF) module. SATR-Blocks form each stage; strided convolutions downsample features and increase channels between stages. A PAN-FPN fuses the backbone’s P3, P4, and P5 features before the detection head. Each SATR-Block comprises MGAM, MTRP, and MTCP.

After layer normalization, MGAM uses residual multiscale aggregation to produce a shared feature with enhanced channel and spatial saliency. This feature feeds the router, feature-projection branch, and MTCP. The router generates a one-to-one macro-region permutation for each of the four channel groups. Each projected group retains all spatial positions, is reordered according to its assigned permutation, processed by an independent Mamba branch, and then restored to the original layout by the inverse permutation before concatenation. MTCP extracts macro structure from the shared feature to generate a channel-wise gate. After the restored group features are modulated by Mamba’s original z gate, MTCP further calibrates them to suppress redundant and non-target responses. Output projection, residual fusion with the block input, and an FFN residual branch yield the block output.

2.2. Multi-Scale Gaussian Attention Modulation

Aggregation into $G \times G$ macro nodes may dilute local saliency; therefore, MGAM enhances local saliency before routing. For $\mathcal{K} = \{1, 3, 5, 7\}$, let $F_r = C_r(X)$, where C_r is an $r \times r$ convolution. At each scale, spatial centers are selected as the top- k locations with the highest values in the sum of the channel-mean and channel-maximum maps; channel centers are the top- k channels ranked by the sum of global average-pooling and global max-pooling responses. The spatial Gaussian uses standard deviations $W/2$ and $H/2$ along width and height, respectively, while the channel Gaussian uses $C/2$. Each branch sums the k Gaussian probability density functions and min-max normalizes the result per sample to obtain W_r^b . For $b \in \{c, s\}$,

let d denote the mean absolute difference between normalized heat maps; then

$$\hat{X} = X + \sum_{r \in \mathcal{K}} \sum_{b \in \{c, s\}} \alpha_r^b Z_r^b, \quad (1)$$

$$Z_r^b = F_r \odot W_r^b, \quad \alpha_r^b = \frac{d(X, Z_r^b)}{\sum_{j \in \mathcal{K}} d(X, Z_j^b) + 10^{-8}}.$$

The indices r and j enumerate scales, b denotes the channel or spatial branch, and $\odot$ denotes elementwise multiplication. The distance $d(A, B)$ is the spatial mean absolute difference between the channel-averaged maps after per-sample min-max normalization; 10^{-8} prevents division by zero. The residual X preserves the input, while $\hat{X}$ aggregates multi-scale responses and serves as the input to the router, the MTRP feature-projection branch, and MTCP.

2.3. Macro-Topology Routed State Propagation

MTRP uses content-adaptive reordering to reduce irrelevant inter-region adjacencies induced by fixed scans. It decouples macro-topology decisions from fine-grained propagation: the router orders only $N = G^2$ macro nodes, while Mamba processes all $H \times W$ positions. After extracting macro nodes from $\hat{X}$ and encoding their context, the routing network produces the logit matrix for group g :

$$L_g = f_g \left(\mathcal{T} \left(\psi(\hat{X}) + E \right) \right) \in \mathbb{R}^{N \times N}, \quad (2)$$

where $g \in \{1, 2, 3, 4\}$, ψ extracts macro nodes, $\mathcal{T}$ encodes their context, E is a learnable positional encoding, and f_g is the g -th predictor head. A positive logit scale and 20 log-domain Sinkhorn iterations with alternating row and column normalization [23] at $\tau = 0.1$ yield an approximately doubly stochastic matrix Q_g . A

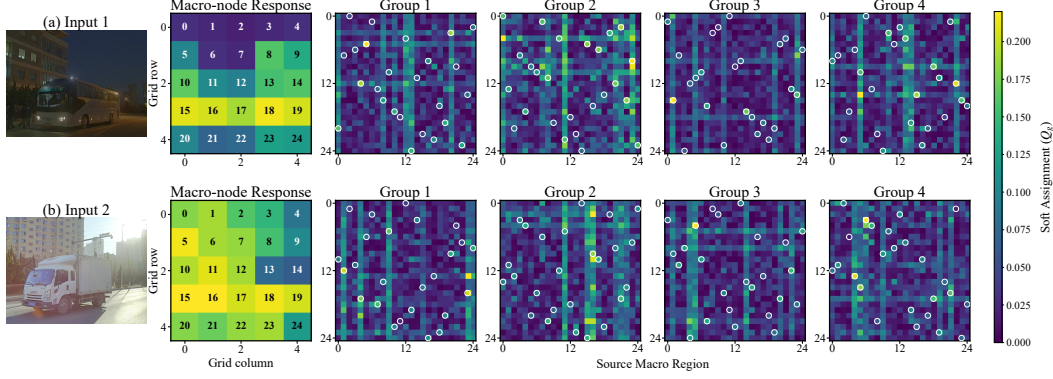


Fig. 2. Input-conditioned, group-specific routing in MTRP. Each row shows one scene’s macro-node response and four group-wise Sinkhorn assignments Q_g ; white circles mark the Hungarian hard permutations.

maximum-affinity one-to-one matching step together with a straight-through estimator enables discrete routing in the forward pass while allowing continuous gradient propagation in the backward pass:

$$Q_g = \exp \left[\text{Sinkhorn}_{\log}^{(20)} \left(\frac{\text{softplus}(\gamma) L_g}{\tau} \right) \right], \quad \tau = 0.1, \quad (3)$$

$$P_g = \arg \max_{P \in \Pi_N} \langle P, Q_g \rangle_F, \quad \bar{P}_g = P_g - \text{sg}(Q_g) + Q_g. \quad (4)$$

Here, P_g is the hard Hungarian permutation; $\bar{P}_g = P_g - \text{sg}(Q_g) + Q_g$ is its straight-through counterpart, equal to P_g in the forward pass but differentiable through Q_g in the backward pass. Here, $\text{softplus}(\gamma)$ provides a positive logit-scaling factor; τ denotes the temperature; P is a candidate permutation; $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product; and Π_N is the set of $N \times N$ permutation matrices. Each routing position matches exactly one source region, and vice versa. For group g , $\bar{P}_g^\top$ reorders regions without changing intra-region order. The reordered group feature is unfolded, processed by an independent Mamba branch, and restored with $\bar{P}_g$:

$$Y_g = \mathcal{R}^{-1} \left\{ \mathcal{M}_g \left[\mathcal{R} \left(B_g \bar{P}_g^\top \right) \right] \right\} \bar{P}_g. \quad (5)$$

In (5), Y_g is the restored output, B_g is its channel subspace, $\mathcal{R}$ and $\mathcal{R}^{-1}$ are the unfolding and inverse transforms, and $\mathcal{M}_g$ is the group-specific Mamba module. The groups can learn distinct input-conditioned orders, thereby inducing different propagation paths while retaining every position.

Fig. 2 shows macro-node response maps for two scenes and the four groups’ Sinkhorn assignments Q_g and Hungarian permutations P_g . Node indices 0–24 follow row-major order. Rows correspond to routing positions, columns to source regions, colors to soft-assignment strengths, and white hollow circles to hard matches. The presence of exactly one circle in each row and column confirms one-to-one assignment without repeated region visits.

2.4. Macro-Topology Cross-Path Gate

Using the pre-routing feature $\hat{X}$, MTCP constructs a shared structural gate to calibrate the restored groups. Channel averaging followed by $G \times G$ adaptive max pooling produces a macro state M . We model M as a two-dimensional field using direction-aware transport with isotropic diffusion:

$$\frac{\partial u}{\partial t} = \beta_q(u) \odot \partial_q u + \lambda \nabla^2 u, \quad q \in \mathcal{D}. \quad (6)$$

Here, $u(0) = M$, $q \in \mathcal{D} = \{x^+, x^-, y^+, y^-\}$, ∂_q is the directional derivative, and ∇^2 is the isotropic Laplacian. A 1×1 convolution followed by a sigmoid produces $\beta_q(u)$, while λ controls diffusion. The implementation evaluates β_q at M and uses one update with $\lambda = 0.1$:

$$U_q = M + \beta_q(M) \odot D_q M + 0.1 \Delta M. \quad (7)$$

The descriptor U_q uses the neighbor-minus-center difference operator D_q with replicate padding and the four-neighbor Laplacian Δ to approximate ∂_q and ∇^2 , respectively. Path-specific Linear-GELU-LayerNorm projections ϕ_q are applied separately to the four descriptors U_q , $q \in \mathcal{D}$. The four projected descriptors are concatenated and passed through two shared residual-mixing steps before an elementwise sigmoid produces the channel gate

$$G_c = \sigma \left(F_{\text{mix}}^{(2)} [\text{Concat}_{q \in \mathcal{D}} \phi_q(U_q)] \right). \quad (8)$$

Here, $F_{\text{mix}}^{(2)}$ denotes two residual applications of the same Linear-SiLU-LayerNorm mixer, each adding 0.5 times its output to its input. The gate G_c is broadcast spatially to calibrate the restored groups.

3. EXPERIMENTS

3.1. Experimental Settings

We use visible-light images from M3FD [24] and MFAD [25], resized to 640×640 pixels. SATR-SSM is implemented in PyTorch 2.4.1 and trained on a single NVIDIA GeForce RTX 2080 Ti (11 GB) using a consistent training configuration across all experiments. We report precision, recall, mAP_{50} , $\text{mAP}_{50:95}$, and parameter count. All experiments use $G = 5$; MGAM uses $k = 6$.

3.2. Comparison Experiments

For each baseline, we use settings from its paper or official implementation when available and evaluate all models on the same hardware. Tables 1 and 2 report M3FD and MFAD results. On M3FD, SATR-SSM reaches an mAP_{50} of 0.706 and an $\text{mAP}_{50:95}$ of 0.435 with 3.72M parameters, outperforming the YOLO s-series baselines, each of which has approximately 9M parameters. This result indicates effective feature modeling with a smaller parameter budget. Although Mamba-ViT-YOLO has only 1.80M parameters, it scores lower on both mAP metrics.

Table 1. Comparison on M3FD. Best values are bold; second-best values are underlined.

Method	P	R	mAP ₅₀	mAP _{50:95}	Params (M)
YOLO11s [26]	0.800	0.590	0.671	0.421	9.42
YOLO12s [27]	0.786	0.602	0.674	0.417	9.23
YOLO13s [28]	0.773	0.566	0.640	0.417	9.00
YOLO26s [29]	0.773	0.617	<u>0.692</u>	0.423	9.45
Hyper-YOLO [30]	<u>0.802</u>	0.590	<u>0.669</u>	0.422	2.04
YOLO-Master [31]	0.793	<u>0.621</u>	0.691	<u>0.432</u>	2.80
LSNet [32]	0.720	0.536	0.631	0.357	19.40
Mamba-YOLO-T [18]	0.789	0.597	0.679	0.421	5.99
Mamba-ViT-YOLO [19]	0.794	0.588	0.667	0.397	1.80
SATR-SSM (Ours)	0.819	0.626	0.706	0.435	3.72

Table 3. Component ablation on M3FD; a checkmark denotes an enabled module.

MTRP	MGAM	MTCP	mAP ₅₀	mAP _{50:95}	Params (M)
—	—	—	0.583	0.355	2.58
✓	—	—	0.607	0.378	3.45
✓	✓	—	<u>0.623</u>	<u>0.385</u>	3.60
✓	—	✓	0.610	0.377	3.55
✓	✓	✓	0.706	0.435	3.72

On MFAD, SATR-SSM also achieves the highest mAP₅₀. Hyper-YOLO and Mamba-ViT-YOLO use fewer parameters than SATR-SSM but score lower on both mAP metrics. YOLO-Master has fewer parameters and a slightly higher mAP_{50:95}; SATR-SSM has a slightly higher mAP₅₀. Thus, the relative advantage of the two models depends on the metric considered.

3.3. Ablation Experiments

Table 3 evaluates different component combinations. Adding MTRP raises mAP₅₀ from 0.583 to 0.607 and mAP_{50:95} from 0.355 to 0.378. Adding MGAM further raises them to 0.623 and 0.385, whereas MTRP+MTCP reaches 0.610 and 0.377. The full configuration yields the highest values, 0.706 and 0.435, suggesting that MGAM and MTCP complement MTRP.

Fig. 3 shows SATR-Block responses in three scenes. MGAM strengthens target-edge responses and local saliency while suppressing background interference. MTRP propagates states along content-adaptive macro-region sequences and restores the spatial layout. Using the MGAM-enhanced feature as a structural reference, MTCP calibrates the restored groups to suppress non-target responses and preserve target structure. These stage-wise changes align with the ablation results in Table 3.

3.4. Visualization Analysis

Fig. 4 compares M3FD visible-light detections across scenes featuring dense targets, low illumination, or distant small targets. Compared with YOLO11s and Hyper-YOLO, SATR-SSM produces detections that agree more closely with the ground truth, with fewer false positives and misses in the yellow dashed regions, more complete detection of adjacent targets and targets at different scales, and better-aligned bounding boxes.

Table 2. Comparison on MFAD. Best values are bold; second-best values are underlined.

Method	P	R	mAP ₅₀	mAP _{50:95}	Params (M)
YOLO11n [26]	0.769	0.528	0.607	0.391	2.58
YOLO12n [27]	0.751	0.512	0.591	0.383	2.51
YOLO13n [28]	0.766	0.527	0.611	0.397	2.45
YOLO26n [29]	0.765	0.556	0.639	0.404	2.38
Hyper-YOLO [30]	0.758	0.547	0.627	0.409	2.04
YOLO-Master [31]	<u>0.770</u>	<u>0.566</u>	<u>0.650</u>	0.430	2.80
LSNet [32]	0.742	0.516	0.612	0.376	19.40
Mamba-ViT-YOLO [19]	0.751	0.534	0.610	0.384	1.80
SATR-SSM (Ours)	0.783	0.578	0.655	<u>0.422</u>	3.72

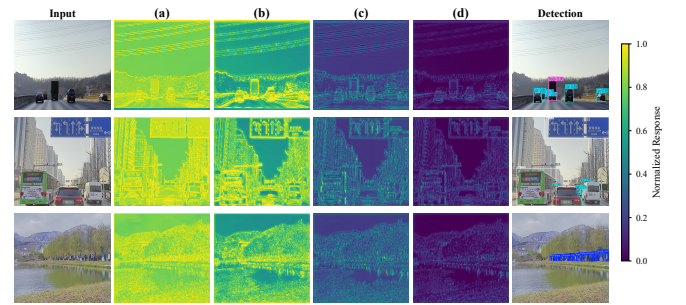


Fig. 3. Stage-wise responses of SATR-SSM. Columns show the input image, SATR-Block input (a), MGAM output (b), MTRP output (c), MTCP output (d), and final detection; rows show low-light small targets, scale variation, and dense, complex backgrounds.



Fig. 4. M3FD visible-light detections. Green, red, and blue boxes denote correct detections, false positives, and misses.

4. CONCLUSION

This paper presents SATR-SSM, which decouples topology selection from the resolution of state propagation. MTRP learns input-conditioned, group-specific one-to-one macro-region permutations while retaining all spatial positions for group-wise Mamba propagation. MGAM provides pre-routing saliency enhancement, while MTCP provides shared gating conditioned on macro-level structure. Experiments demonstrate a competitive trade-off between detection accuracy and model size. Future work will focus on reducing computational cost and investigating routing sparsity, macro-grid sizes, and permutation constraints.

5. REFERENCES

- [1] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [2] Tri Dao and Albert Gu, “Transformers are ssms: Generalized models and efficient algorithms through structured state space duality,” *arXiv preprint arXiv:2405.21060*, 2024.
- [3] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang, “Vision mamba: Efficient visual representation learning with bidirectional state space model,” *arXiv preprint arXiv:2401.09417*, 2024.
- [4] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, Jianbin Jiao, and Yunfan Liu, “Vmamba: Visual state space model,” *Advances in neural information processing systems*, vol. 37, pp. 103031–103063, 2024.
- [5] Tao Huang, Xiaohuan Pei, Shan You, Fei Wang, Chen Qian, and Chang Xu, “Localmamba: Visual state space model with windowed selective scan,” in *European conference on computer vision*. Springer, 2024, pp. 12–22.
- [6] Shufan Li, Harkanwar Singh, and Aditya Grover, “Mamba-nd: Selective state space modeling for multi-dimensional data,” in *European Conference on Computer Vision*. Springer, 2024, pp. 75–92.
- [7] Leiye Liu, Miao Zhang, Jihao Yin, Tingwei Liu, Wei Ji, Yongri Piao, and Huchuan Lu, “Defmamba: Deformable visual state space model,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 8838–8847.
- [8] Tanzhe Li, Caoshuo Li, Jiayi Lyu, Hongjuan Pei, Baochang Zhang, Taisong Jin, and Rongrong Ji, “Damamba: Vision state space model with dynamic adaptive scan,” *Advances in Neural Information Processing Systems*, vol. 38, pp. 39777–39799, 2026.
- [9] Kun Zhou, Xinyu Lin, and Jiangbo Lu, “Tsp-mamba: The travelling salesman problem meets mamba for image super-resolution and beyond,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 28134–28143.
- [10] Zheyu Zhang, Yayuan Lu, Feipeng Ma, Yueyi Zhang, Huanjing Yue, and Xiaoyan Sun, “Incomplete multi-modal brain tumor segmentation via learnable sorting state space model,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 25982–25992.
- [11] Xiaohuan Pei, Tao Huang, and Chang Xu, “Efficientvmamba: Atrous selective scan for light weight visual mamba,” in *Proceedings of the AAAI conference on artificial intelligence*, 2025, vol. 39, pp. 6443–6451.
- [12] Ali Hatamizadeh and Jan Kautz, “Mambavision: A hybrid mamba-transformer vision backbone,” in *2025 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. IEEE, 2025, pp. 25261–25270.
- [13] Yuheng Shi, Mingjia Li, Mingjing Dong, and Chang Xu, “Vssd: Vision mamba with non-causal state space duality,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 10819–10829.
- [14] Chaodong Xiao, Minghan Li, Deyu Meng, Lei Zhang, et al., “Spatial-mamba: Effective visual state space models via structure-aware state fusion,” in *International Conference on Learning Representations*, 2025, vol. 2025, pp. 73777–73795.
- [15] Xiaowen Ma, Zhenliang Ni, and Xinghao Chen, “Tinyvim: Frequency decoupling for tiny hybrid vision mamba,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 23519–23529.
- [16] Fei Xie, Zhongdao Wang, Weijia Zhang, and Chao Ma, “Pv-mamba: Parallelizing vision mamba via dynamic state aggregation,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 10218–10228.
- [17] Weihao Yu and Xinchao Wang, “Mambaout: Do we really need mamba for vision?,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 4484–4496.
- [18] Zeyu Wang, Chen Li, Huiying Xu, Xinzong Zhu, and Hongbo Li, “Mamba yolo: A simple baseline for object detection with state space model,” in *Proceedings of the AAAI conference on artificial intelligence*, 2025, vol. 39, pp. 8205–8213.
- [19] Ze Song, Yuhai Wang, Shuobo Xu, Peng Wang, and Lele Liu, “Lightweight vehicle detection based on mamba_vit,” *Sensors*, vol. 24, no. 22, pp. 7138, 2024.
- [20] Long Chen, Hui Wang, Man Xu, Zexuan Li, and Zizhu Fan, “Akcmamba-yolo: Selective state space models for real-time object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 4438–4447.
- [21] Ziyao Zhang, Qiankun Ma, Tong Zhang, Jie Chen, Hairong Zheng, and Wen Gao, “Switch-umamba: Dynamic scanning vision mamba unet for medical image segmentation,” *Medical Image Analysis*, p. 103792, 2025.
- [22] Abdelrahman Shaker, Syed Talal Wasim, Salman Khan, Juergen Gall, and Fahad Shahbaz Khan, “Groupmamba: Efficient group-based visual state space model,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 14912–14922.
- [23] Marco Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” *Advances in neural information processing systems*, vol. 26, 2013.
- [24] Jinyuan Liu, Xin Fan, Zhanbo Huang, Guanyao Wu, Risheng Liu, Wei Zhong, and Zhongxuan Luo, “Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection,” in *2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. IEEE, 2022, pp. 5792–5801.
- [25] Ke Hu, Yudong He, Yuan Li, Jiayu Zhao, Song Chen, and Yi Kang, “Ei²det: Edge-guided illumination-aware interactive learning for visible-infrared object detection,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 7, pp. 7101–7115, 2025.
- [26] Glenn Jocher and Jing Qiu, “Ultralytics yolo11,” 2024.
- [27] Yunjie Tian, Qixiang Ye, and David Doermann, “Yolov12: Attention-centric real-time object detectors,” *Advances in neural information processing systems*, vol. 38, pp. 78433–78457, 2026.
- [28] Mengqi Lei, Siqu Li, Yihong Wu, Han Hu, You Zhou, Xinhui Zheng, Guiguang Ding, Shaoyi Du, Zongze Wu, and Yue Gao, “Yolov13: Real-time object detection with hypergraph-enhanced adaptive visual perception,” *arXiv preprint arXiv:2506.17733*, 2025.
- [29] Glenn Jocher, Jing Qiu, Mengyu Liu, Shuai Lyu, Fatih Cagatay Akyon, and Muhammet Esat Kalfaoglu, “Ultralytics yolo26: unified real-time end-to-end vision models,” *arXiv preprint arXiv:2606.03748*, 2026.
- [30] Yifan Feng, Jiangang Huang, Shaoyi Du, Shihui Ying, Jun-Hai Yong, Yipeng Li, Guiguang Ding, Rongrong Ji, and Yue Gao, “Hyper-yolo: When visual object detection meets hypergraph computation,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 47, no. 4, pp. 2388–2401, 2024.
- [31] Xu Lin, Jinlong Peng, Zhenye Gan, Jiawen Zhu, and Jun Liu, “Yolo-master: Moe-accelerated with specialized transformers for enhanced real-time detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 18440–18449.
- [32] Ao Wang, Hui Chen, Zijia Lin, Jungong Han, and Guiguang Ding, “Lsnet: See large, focus small,” in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2025, pp. 9718–9729.